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Why Blockchain?

Start with the problem, not the technology.





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In this module: We learn how to decide whether blockchain is actually justified — by examining trust, shared control, verification needs, and the trade-offs of using it.

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Could ≠ Should

A system can use blockchain. That does not mean blockchain is the right architectural choice.



Key Takeaway: Technical feasibility is not architectural justification.
Start with the problem and its requirements before deciding whether blockchain belongs in the solution.

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Start With the Problem

Before choosing a technology, understand the problem, the participants, and the trust relationships between them.

Participants
Who creates, updates, or uses the shared information?



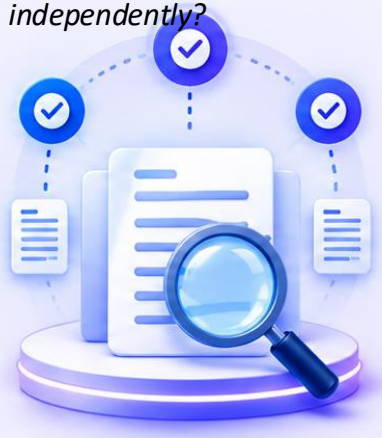
Shared State
What information must multiple participants share?



Trust
Who is trusted to control or change that information?



Verification
Who needs to verify records independently?



Key Takeaway: The first question is not **"Can blockchain solve this?"** but **"What problem and trust model are we trying to support?"**

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Is There an Acceptable Trusted Authority?

If all participants accept one organization as the authority, a centralized system may already solve the problem.

Multiple Participants
Several parties need access to the same records.



Trusted Authority Exists
All participants accept one organization to manage the records.



Centralized System May Be Enough

No Commonly Trusted Authority
Participants cannot — or should not have to — rely on one party for control.



Consider Shared Control

Key Takeaway: If a trusted authority is acceptable, blockchain may add complexity without solving a real trust problem.

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Who Needs to Share Control?

Decentralization matters when multiple parties need to participate without giving one of them unilateral control.

ONE PARTY CONTROLS THE RECORD

Centralized Control



CONTROL IS SHARED

Independent Participants



Key Takeaway: Blockchain becomes relevant when shared control is a requirement — not simply because multiple participants are involved.

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What Must Be Independently Verifiable?

A shared system becomes more valuable when participants need to verify records without relying on the party that created them.



Key Takeaway: Independent verification reduces the need to trust claims about a record — but it does not prove that the record itself is true.

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Capabilities Come With Trade-offs

Blockchain can provide useful properties – but they come with cost and constraints.

CAPABILITIES

- Shared control
- Independent verification
- Tamper-evident records



TRADE-OFFS

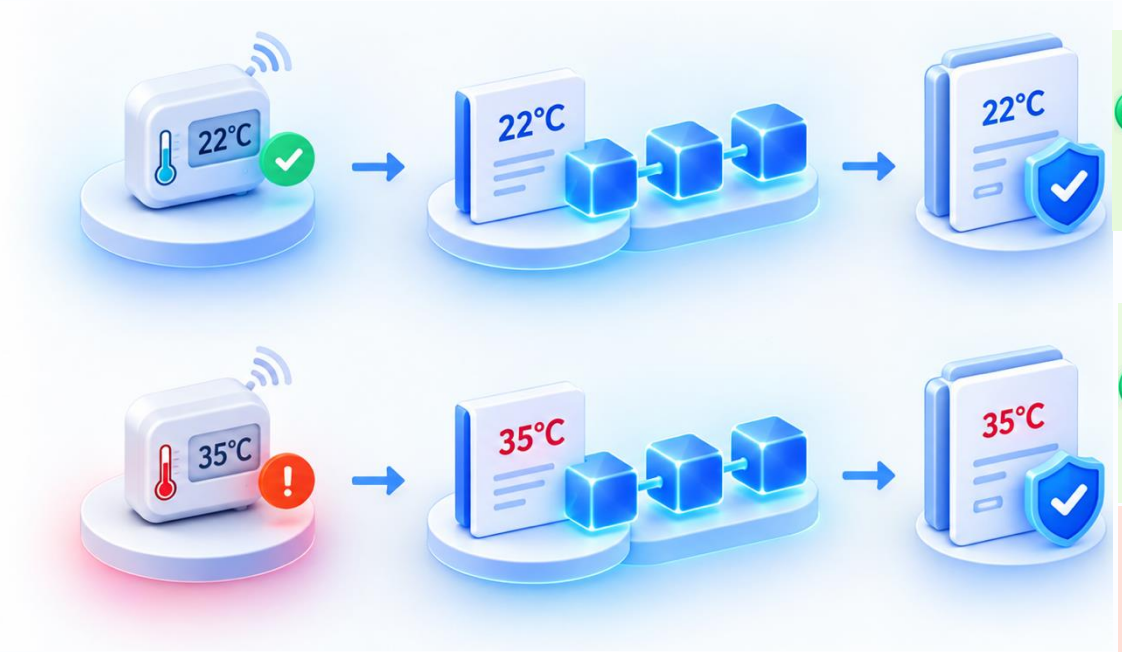
- Complexity
- Cost & performance
- Governance burden

Key Takeaway: The question is not whether blockchain provides useful capabilities — but whether those capabilities justify the trade-offs.

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Integrity Is Not Truth

Blockchain can protect a record from undetected changes — but it cannot guarantee that the original information was correct.



Integrity preserved

The record is unchanged and verifiable

Integrity preserved

The record is unchanged and verifiable

But the claim is still wrong.

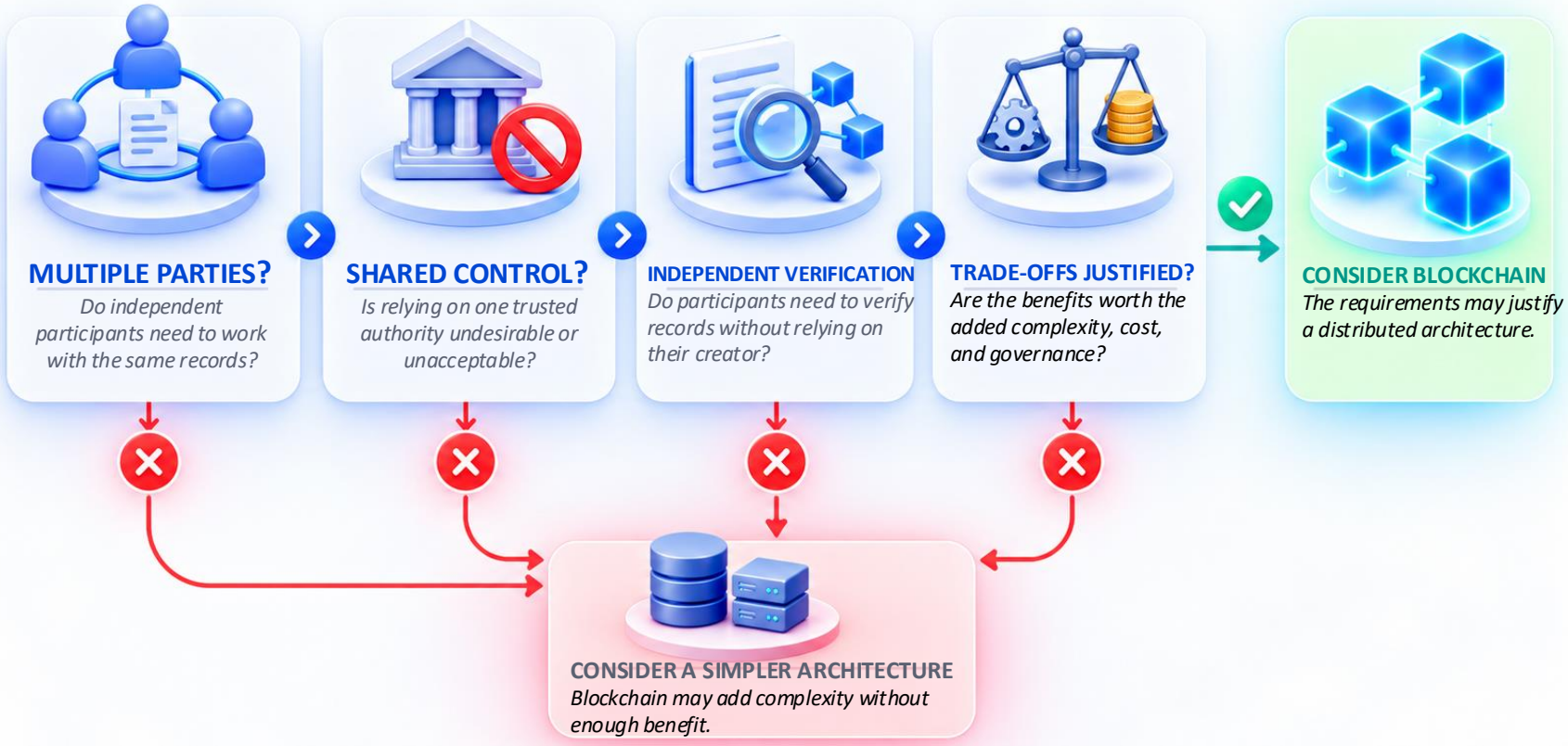
The original information was incorrect

Key Takeaway: An immutable false claim is still a false claim.

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A Framework for Deciding

Before choosing blockchain, ask a sequence of questions about the problem — not the technology.



Key Takeaway: Blockchain is justified by a combination of requirements, not by a single feature.

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Three Valid Conclusions

A good decision process does not always end with “use blockchain.” Different requirements can lead to different architectures.



1 CENTRALIZED SYSTEM

When there is an acceptable trusted authority and no need for shared control

BEST FIT WHEN

A single organization can manage the records efficiently and participants accept its authority.

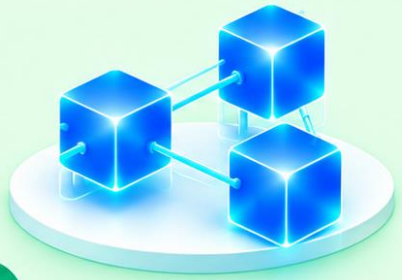


2 HYBRID APPROACH

When only specific records or functions need shared control or independent verification.

BEST FIT WHEN

Blockchain adds value to part of the system, while other data and functions can remain centralized.



3 BLOCKCHAIN

When shared control, independent verification, and tamper-evidence justify the added trade-offs.

BEST FIT WHEN

No single party should control the shared record, and verifiability across independent participants matters.

Key Takeaway: The goal is not to choose blockchain — it is to choose the architecture that best fits the problem.

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Apply the Framework: Academic Credentials

Suppose universities, employers, and learners need to issue, hold, and verify academic credentials across organizational boundaries.



1 MULTIPLE PARTIES?	2 SHARED CONTROL?	3 INDEPENDENT VERIFICATION?	4 TRADE-OFFS JUSTIFIED?	5 HYBRID APPROACH
YES	PARTLY	YES	DEPENDS	LIKELY FIT
Universities, learners, employers, and other institutions participate in the credential ecosystem.	Each university controls what it issues, but no single organization needs to control the entire ecosystem.	A verifier should be able to check a credential without contacting the issuing university every time.	Blockchain may add value for proofs, issuer registries, or revocation status without storing the credential itself on-chain.	Use blockchain where independent verification adds value, while keeping personal data and credential contents off-chain.

Key Takeaway: Blockchain becomes relevant because verification crosses organizational boundaries — not simply because the records are digital.

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What Blockchain Does — and Does Not — Solve

Blockchain provides specific trust and verification capabilities. It does not remove the need for good data, governance, or system design.

BLOCKCHAIN CAN HELP WITH



1 — SHARED CONTROL
Reduces dependence on a single controlling party



2 — INDEPENDENT VERIFICATION
Allows participants to verify records without relying only on their creator.



3 — TAMPER-EVIDENCE
Makes changes to recorded information detectable.



4 — TRACEABILITY
Preserves a verifiable history of recorded actions.



BLOCKCHAIN DOES NOT GUARANTEE



1 — TRUTH OF INPUT
Incorrect information can still be recorded immutably.



2 — GOOD GOVERNANCE
Rules, roles, and responsibilities still need to be designed.



3 — PRIVACY BY DEFAULT
Sensitive information still requires careful handling.



4 — A BETTER ARCHITECTURE
Blockchain can add unnecessary complexity when simpler systems are sufficient.

Key Takeaway: Blockchain changes how trust is distributed and verified — it does not eliminate the need for trust, judgment, or good system design.

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Sometimes the Answer is: Not Enough Information

Choosing an architecture too early can be as problematic as choosing the wrong one.



Key Takeaway: "Not enough information" is a valid engineering conclusion — choosing a technology before the requirements are clear is premature.

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What to Remember

Five principles for deciding when blockchain is — and is not — the right architectural choice.

1 START WITH THE PROBLEM

Identify the trust and coordination problem before choosing the technology.

2 LOOK FOR SHARED CONTROL

Blockchain becomes relevant when independent participants need to coordinate without relying on one controlling authority.

3 ASK WHAT MUST BE VERIFIED

Use blockchain where participants need to verify records independently across organizational boundaries.

4 REMEMBER THE LIMITS

Blockchain can preserve integrity and traceability, but it cannot guarantee truth, privacy, or good governance.

5 JUSTIFY THE TRADE-OFFS

The benefits must justify the added complexity, cost, performance constraints, and governance burden.

Key Takeaway: Use blockchain because the problem requires its properties — not because the technology is available.

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Visual Lesson Complete!

You now have a framework for deciding when blockchain is useful, when a simpler architecture may be sufficient, and when more information is needed before making a choice.



Next Steps

1. **Apply the decision framework**
   Use the questions from this module to evaluate a real-world scenario.
2. **Explore the LM02 activities**
   Test when shared control and independent verification justify blockchain.
3. **Complete the assessment**
   Check your understanding of the key concepts from LM02.

Continue Your Learning Path
Continue with the LM02 activities in Web3Edu.

Key Takeaway: The best blockchain decision is sometimes to use blockchain — and sometimes not to.

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